Product demand sensitivity and the corporate diversification discount

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Abstract

We consider consumer responsiveness to changes in the macroeconomic environment (i.e. product demand sensitivity) to be a systematic industry characteristic useful to studying how industrial diversification adds value. We argue that diversified firms, with the advantages of their internal capital market and the imperfectly correlated cash flows of their segments, will perform better than focused firms when the demand sensitivity of a product increases. Our empirical evidence reveals a significant diversification premium associated with an increase in sensitivity. Moreover, such premium predominantly exists during recessionary periods, disappearing in the presence of low coinsurance and inefficient use of the internal capital market. Our results are robust to alternative measures of sensitivity and performance metrics, different empirical model specifications, and to the concern of possibly biased estimations associated with the sample of diversified and focused firms.

Introduction

How does conglomeration affect value? The dominant view within conventional literature is that corporate diversification usually destroys value (e.g., Lang and Stulz, 1994; Berger and Ofek, 1995). This is, to some extent, puzzling since the economic benefits of diversification should be nontrivial. Considering the systematic channels through which conglomerates might create value, recent evidence suggests that diversified firms, in fact, can create more value than their single-segment counterparts in certain economic conditions (e.g., Dimitrov and Tice, 2006; Santalo and Becerra, 2008; Gopalan and Xie, 2011; Kuppuswamy and Villalonga, 2016).[1] In this paper, we argue that product demand sensitivity, which measures the sensitivity of an industry's revenue generation capacity to macroeconomic performance, is a systematic industry characteristic that can create a diversification premium for conglomerates.

Sensitivity of a firm's performance to changes in the macroeconomic environment may be heterogeneous across industries. Importantly, while sensitivity may vary across industries, so too may the level of sensitivity vary over time within an industry. For example, consumers could become more sensitive to the prospects of their earnings conditions, revealed by the macroeconomic environment, and be less loyal to pricey alternatives in their decision of purchasing automobiles if alternative transportation systems became more accessible, or, alternatively, if there is a sharp increase in fuel cost. In other words, factors such as technological advancements and structural changes in economic activities could make certain industries more sensitive to economic cycles than others.[2] Such product demand sensitivity, or the sensitivity of a product's performance to economic cycles, can create systematic advantages for diversified firms over their single-segment rivals.

Consider an economy in an expansionary state in which highly sensitive products tend to experience higher growth in demand. To successfully utilize a market with growing demand requires immediate action by the companies. In addressing such opportunities, diversified firms can use the internal capital market (e.g., Stein, 1997; Matsusaka and Nanda, 2002; Mathews and Robinson, 2008) as an option unavailable to their specialized counterparts. Alternatively, in a shrinking economy, highly sensitive products would likely experience falling demand at a greater magnitude than less sensitive products. In such an economy, diversified firms, with the internal capital market and imperfectly correlated cash flows of business units, are more likely to outperform their single-segment counterparts, as the findings of prior literature suggest (e.g., Dimitrov and Tice, 2006; Gopalan and Xie, 2011; Kuppuswamy and Villalonga, 2016).

We predict that, on average, any increase in the sensitivity of product demand will create systematic opportunities for a diversification premium. For our empirical tests, we create a measure of demand sensitivity for each industry for each year using the beta coefficient from a rolling regression of the log difference of annual 4-digit SIC industry sales on the log difference of annual aggregate sales across industries, using the past 10–20 years of annual data.

Since a diversified firm operates in more than one 4-digit SIC industry, we calculate firm-level sensitivity as the weighted average of sensitivities across segment-industries, where weight is the fraction of a segment's share in the aggregate sales of an industry.[3]

In our multivariate analyses, we document a statistically significant 16%–24% diversification premium associated with an increase in demand sensitivity. In examining the impact of sensitivity on performance during recessionary periods, we find that the subsample of firms with at least 50% of segments operating in high sensitivity industries enjoy a diversification premium associated with recessions of as much as 16.7%. In our further analyses, we focus on how sensitivity impacts performance when firms change their diversification status. We do not find any evidence of a premium associated with the rise in sensitivity when a diversified firm refocuses on a single industry. On the other hand, when focused firms become diversified, we find evidence of a significant diversification premium associated with the increase in sensitivity.

Additionally, we test our results directly in the context of coinsurance and internal capital market, which are the two important factors that help diversified firms realize any advantages associated with the change in industry sensitivity. We do not find any evidence of a significant diversification premium associated with the increase in sensitivity in the case of firms that exhibit inefficient allocation of capital across their segments. In fact, conglomerates that have both low coinsurance and an inefficient internal capital market tend to experience a diversification discount when industry sensitivity increases. This suggests that the advantages that come with the increase in sensitivity cannot be realized in the presence of inefficient use of resources and high cash flow correlation across segments.

In further tests, we account for variables that capture the market dominance of diversified versus specialized firms, as Santalo and Beccera (2008) find evidence of a diversification premium for firms operating in industries where diversified firms hold a substantial market share. After controlling for the number of specialized competitors in our regression model, we find a diversification premium that is similar in magnitude to that of our earlier tests. In a separate test, we find that our main findings persist when controlling for the market share of specialized competitors as well.

In our empirical specifications, we employ several approaches to address the possibility of biased estimations using a sample of diversified and focused firms (Campa and Kedia, 2002; Villalonga, 2004a). First, since the decision of diversification might not be random and is likely to be influenced by firm-specific characteristics, we use firm fixed effects in our basic specification models to address this issue. Second, since the changing of the diversification status of a firm is not a frequent event, models using fixed effects might lose some explanatory power in the presence of a little within variation. Therefore, in addition to the use of the fixed effects model, we use a modified version of the random effects model in our robustness tests.[4] Third, following Campa and Kedia (2002), we use the Heckman (1979) model to account for endgogeneity associated with diversification decisions, and additionally, we use two instruments for diversification in a two-stage model.

In further robustness tests, we use an alternative measure of weighted average demand sensitvity; we use GDP as the indicator of the macroeconomic environment and find that our results continue to hold. Our main measure of performance is the excess value of sales, which might lead to a concern of a potential mechanical relationship between sales-based performance and sales-based sensitvity. Thus, we use excess value of assets as an alternative measure of performance and find a signficant diversification premium associated with the increase in sensitvity.

Finally, in our robustness tests, we check how diversification can generate value when companies are involved in highly sensitive acquisitions. In doing so, we identify diversifying acquisitions with bidders whose 4-digit SIC of operations is different than the target's 4-digit SIC. Then we identify diversifying bidders who make sensitive acquisitions, which are M&As where the target's industry beta, as the measure of sensitivity, is higher than the bidder's beta and the bidder experiences an abnormal increase in firm-level sensitivty in the M&A year. In our cross-sectional regressions, using seven-day cumulative abnormal returns of bidders surrounding the announcement date as the dependent variable, we find that the diversifying acquirers tend to achieve as much as 5% additional value when the targets are from highly sensitive industries.

Our results contribute to multiple streams of literature in finance and economics. First, while prior literature focuses primarily on identifying factors related to cash flows that can explain how diversification affects value,[5] we explore whether there is any systematic channel that influences the intrinsic value of conglomerates. Early literature argues that diversified firms can access benefits associated with the debt coinsurance effect (Lewellen, 1971), efficient internal capital markets (e.g., Williamson, 1970, 1975; Stein, 1997), economies of scope (Teece, 1980, 1982), and market dominance (e.g., Tirole, 1995). Each can result in a premium, although evidence from corporate diversification discount literature may suggest otherwise. A few recent papers provide evidence that there are systematic channels through which conglomerates can generate a diversification premium. For example, diversified firms can perform relatively better than their single-segment counterparts during recessionary periods (Hovakimian, 2011; Kuppuswamy and Villalonga, 2016), when corresponding industries become distressed (Gopalan and Xie, 2011), and when the single-segment firms hold a relatively smaller market share of their industries (Santalo and Becerra, 2008). Our paper contributes to this line of literature on corporate diversification by showing that economic diversification can add value when the performance of the industry becomes more sensitive to the performance of the economy.

Second, our paper contributes to the industrial organization literature by providing evidence for the argument that industry structure matters in creating firm value. Even though prior literature considers industry characteristics as important factors that might influence the diversification decision (e.g., Campa and Kedia, 2002; Villalonga, 2004a; Santalo and Becerra, 2008), existing literature does not provide adequate evidence in showing how variation within certain industry structures can affect the diversification-performance relationship. We complement previous empirical findings by studying a fundamental industry characteristic, the industry sensitivity to macroeconomic fluctuations, and find that diversified firms have natural advantages over their specialized counterparts in absorbing the shock associated with a change in sensitivity.[6] In other words, economic diversification makes a firm efficient with regards to shock absorption, which arises when an industry becomes more sensitive to macroeconomic movement, thus creating value advantages over specialized firms.

Third, prior literature suggests that diversified firms perform better than their specialized counterparts due mainly to market conditions within a bad economy state (e.g., Dimitrov and Tice, 2006; Kuppuswamy and Villalonga, 2016). We present a systematic channel that can generate a diversification premium in a good or bad state of the economy. We argue that in a good state of the economy, an increase in industry sensitivity provides opportunities for diversified firms to utilize their internal capital market to capitalize on opportunities associated with sensitivity increase. On the other hand, in a bad state of the economy, diversified firms can utilize their internal resources to efficiently cope with the challenges associated with high sensitivity industries during such times.

Finally, we contribute to an inadequately addressed issue in the macroeconomics and industrial organization literature that examines industry cyclicality, or sensitivity, at a highly disaggregated level. While prior literature recognizes that durable goods industries are more sensitive than non-durable goods industries (e.g., Lucas, 1977; Bernanke, 1983), few papers apply the concept of industry sensitivity to a more granular level. We derive the measure of sensitivity at the 4-digit SIC level and allow it to vary over time and employ it to examine how sensitivity affects the performance of diversified versus specialized firms. Our results help introduce and appreciate the importance of industry sensitivity in the context of the literature on organizational forms.

The remainder of the paper is organized as follows. Section 2 provides a discussion related to literature and hypothesis; Section 3 discusses data and methodology; Section 4 discusses the empirical results; Section 5 provides a description on the findings from the robustness tests; and Section 6 concludes.

Section snippets

The unsettling feature of the diversification discount

The seminal research of Lang and Stulz (1994) and Berger and Ofek (1995) first show evidence of the phenomenon of a "diversification discount," which is the finding that diversified firms generate a lower value than what is generated by a comparable portfolio of specialized firms. A large body of research has since inquired why such a discount might take place, given that there are multiple benefits associated with a firm's strategy of diversifying into different business segments.

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Sample selection

Our sample starts with firm-years for which the data needed to construct the variables in this paper are available in both COMPUSTAT Industrial Annual and Segment data files for the period 1992–2010. We follow the sample selection criterion as described in Berger and Ofek (1995). We include firm-years that do not have any segments in the financial service or utility industries (SIC codes 6000–6999 and 4900–4999) and that have total annual sales of at least $20 million. We confirm that the

Summary statistics

Table 1 describes the key firm characteristics of the whole sample of firms and of single-segment and multi-segment firms. Both mean and median statistics of *Total assets* indicate that diversified firms tend to be larger than focused firms, and the differences between the two groups are significant at the 1% level. On the other hand, the standard deviation of assets for diversified firms is more than two times greater than that of focused firms, which indicates that the former group has a

Competition structure and weighted average demand sensitivity

As mentioned earlier, Santalo and Becerra (2008) both analytically and empirically examine the implications of the argument that the diversification effect on performance should be heterogeneous across industries. They focus on the underlying competitive forces in industries that affect both diversified and focused firms. They examine the market share of diversified firms relative to focused firms in the corresponding industries and find that the performance of diversified firms is negatively

Conclusion

In this paper, we examine whether and how the exposure of industry performance to the macroeconomic environment, or product demand sensitivity, can influence the performance of corporate diversified firms. We argue that depending on the direction of the economy, product demand sensitivity can provide differing economic opportunities for conglomerates. In an expansionary state of the economy, industries with high sensitivity to macroeconomic performance may experience a higher increase in demand

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